

Year 12 Physics
Sem. 1 Exam 2017

Format

Reading Time: 10 mins

Working Time: 3 hrs

Structure

Section 1: 12 short-answer questions

Section 2: 5 problem-solving questions

Section 3: 2 reading comprehensions

Areas of Study

Motion

- Vectors - one and two-dimensional, components.
- Projectile motion - vertical and horizontal movement.
- Circular motion - vertical, horizontal, conical pendulum, banking.
- Satellites - orbital speed, period, geo-stationary orbit, energy.

Equilibrium

- Turning effect of a force.
- Centre of mass and stability.
- Conditions for equilibrium.
- Forces in structures.

Electricity

- Electric and magnetic field drawings.
- Force on current-carrying wires, isolated charges.
- Motors - calculations, torque, explanation of construction.